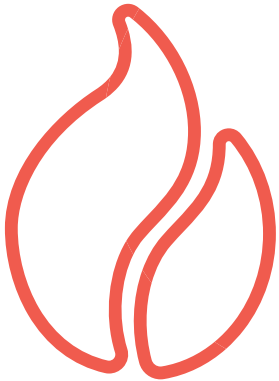


Roasting

Professional AST Guidebook (中文)



Roasting Professional

AST Guidebook V1.0 (中文/Simplified Chinese)

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注意：本指南替代了以前的课程文件。本指南仅适用于获得烘焙技能许可的授权培训师。请不要与其他 AST 或学习者共享此文档。

Note: This guidebook replaces previous curriculum documents. This guidebook is only available to authorized trainers licensed in Roasting. Please do not share this document with other ASTs or learners.

1. 一般信息 General Information

课程信息 Course Information

课程长度：最少 21 小时 Course Length: Minimum 21 hours

通过以下方式花费的时间可以满足最低课程时间 *Minimum course hours are met through time spent in the following ways:*

- AST 可亲自或通过远程学习提供实时指导。
The AST delivering live instruction in-person or through distance learning.
- AST 进行非正式和正式评估。（注意：这不包括学习者参加在线笔试所花费的时间。）
The AST conducting informal and formal assessments. (Note: This does not include time spent by the learner taking the online written exam.)
- 学习者参与实践课程活动以强化基础知识。 The learner engaging in activities that introduce and/or reinforce the theoretical or practical course objectives

课程预备条件：烘焙中级。强烈推荐感官中级和咖啡生豆中级 . Sensory Intermediate and Green Intermediate highly recommended.

笔试信息 Written Exam Information:

在线笔试总题数：30（每项价值 1 分） 80% Total Number of Questions on Online Written Exam: 30 (worth one point each)

在线笔试总时间：32 分钟 Total Time Allowed for Online Written Exam: 32 minutes

及格分数（在线笔试）： Passing Score (Online Written Exam): 80%

实操考核信息： Practical Exam Information:

实操考核总时间：200 分钟 Total Time Allowed for Practical Exam: 200 minutes

及格得分：80% Passing Score (Practical Exam): 80%

2. 课程说明和更新 Course Description and Updates

描述 Description

烘焙高级课程的设计是基于烘焙中级课程中介绍的概念和技能的。学习者将在烘焙曲线发展和感官分析评估方面获得高级技能。本课程深入探讨了广泛的主题，包括在不同和指定的时间限制内进行控制和颜色匹配，烘焙曲线软件的使用和配置，参与褐变反应的分子，烘焙过程中的气体形成，颜色的化学起因以及对溶解度的影响，烘焙瑕疵的视觉识别，拼配，质量控制以及各种生产方案的探索，以帮助满足不同的客户喜好。笔试确认了专业的课程知识，而实践考试则评估了学习者在狭窄的范围内按不同的发展时间目标进行烘焙的能力，目测烘焙颜色，通过杯测识别常见的烘焙瑕疵，并最终区分烘焙的小与大差异烘焙曲线以模拟生产质量控制过程。

The Roasting Professional course is designed to build upon the concepts and skills introduced in the Roasting Intermediate course. Learners will gain advanced skills in profile development and sensory analysis evaluation. This course dives deep into a wide range of topics including control and color matching within different and specified time limits, use and configuration of roast profile software, molecules involved in browning reactions, gas formation during roasting process, chemical causes of color and impact on solubility, visual identification of roasting defects, blending, quality control as well as an exploration of production options to help meet differing customer preferences. A written exam confirms professional I course knowledge while a practical exam assesses the learner's ability to roast to different development time targets within narrow limits, score a roast color visually, identify common roast defects through cupping and finally distinguish between small and large differences in a roast profile to simulate production quality control processes.

先前版本的课程更新 Curriculum Updates from Previous Version

烘焙高级课程已进行了重大更新和修订，以提供清晰度，改进的结构以及与 AST 可以用来支持其提供的培训的知名参考文献的清晰链接。AST 应仔细阅读本节以及第 4 节中的实际课程，以确保其教材是最新的。

The Roasting Professional curriculum has been significantly updated and revised to provide clarity, an improved structure, and clear links to reputable references that ASTs can use to support the training they provide. ASTs should review in detail this section as well as the actual curriculum in Section 4 to ensure their teaching materials are up to date.

课程删除 Deletions to Curriculum:

- 1.01.04 - 不同加工方法的化学性质 Chemical Properties of Different Processing Methods
- 3.01.04 - 失水 Loss of Water
- 4.01.06 - 烘焙速度 Roast Speed
- 6.01.04 – 咖啡豆基因、风土 Bean Genetics, Bean Terroir
- 6.01.08 - 在咖啡中添加牛奶 Addition of Milk to Coffee
- 7.01.03 - 环境，污染，邻居投诉，法规，过滤器选项 Environment, Pollution, Neighbor Complaints, Regulation, Filter Options

课程补充 Additions to Curriculum:

- 3.02.02.03 - 水在热扩散中的作用 Water's Role in Heat Diffusion
- 3.02.03 - 传热和扩散及其对烘焙瑕疵的影响 Heat Transfer and Diffusion and Their Effect on Roast Defects
- 3.07.06 - 烘焙度对清洁和维护的影响 Roast Degree's Effect on Cleaning and Maintenance

课程修订 Revisions to Curriculum:

- 2.0 咖啡烘焙的物理学（以前的课程）-将本节修订并更新为 3.02 咖啡烘焙的热力学。 Physics of Coffee Roasting (previous curriculum) - Section revised and updated to 3.02 Thermodynamics in Coffee Roasting.
- 3.0 豆子的物理变化（以前的课程）-对该栏目进行了修订并更新为 3.03 物理变化。 3.0 Physical Changes in Beans (previous curriculum) - Section revised and updated to 3.03 Physical Changes.
- 4.0 咖啡烘焙化学（以前的课程）-将本节修订并更新为 3.04 化学变化。 4.0 Chemistry of Coffee Roasting (previous curriculum) - Section revised and updated to 3.04 Chemical Changes.
- 1.01.05 丙烯酰胺主题移至 3.04.06。 1.01.05 Acrylamide topic moved to 3.04.06.
- 6.0 商业（以前的课程）-本栏目修订并更新为 3.06 批发和零售商业 6.0 Business (previous curriculum) - Section revised and updated to 3.06 Wholesale and Retail Business
- 6.01.05 组件数和 6.01.07 拼配前/后已移至 3.07 烘焙管理栏目。 6.01.05 Number of Components and 6.01.07 Pre/post blend moved to 3.07 Roastery Management section.
- 7.01.05 工作区管理主题-防火和灭火已删除。主题更新为 3.07.06-烘焙度对清洁和维护的影响 7.01.05 Workspace Management topic - fire prevention and extinguishing removed. Topic updated to 3.07.06 - Roast Degree's Effect on Cleaning and Maintenance

其他增补和修订 Other Additions and Revisions:

- 修改了所有章节，主题和目标的编码 Revised coding for all sections, topics and objectives
- 修订的 AST 注释和大多数主题的参考 Revised AST Notes and detailed references for most topics

3. 书面笔试题按主题分布 Written Exam Question Distribution by Topic

下表列出了有关在线考试问题的关键信息。

题库：这是每个**主题**可以在在线考试期间向学习者提出的问题数量。

提出的问题：这是学习者将在在线考试中针对**每个主题**随机接收的问题数。该数字由创建者确定，目的是确保对课程的每个栏目和主题进行适当加权。

栏目权重：每个栏目标题旁边是该栏目中的问题所代表的全部考试的百分比。

The chart below sets forth key information regarding the online exam questions.

Question Pool: This is the number of questions **per topic** that are available to present to the learner during the online exam.

Questions Presented: This is the number of questions a learner will randomly receive **per topic** during the online exam. This number was determined by the creators for the purpose of ensuring that each section and topic of the course is weighted appropriately.

Section Weighting: Next to each section title is the percentage of the total exam represented by the questions in that section.

There are some topics/objectives that are not tested during the written or practical exam. The trainer should still address these topics/objectives as the information will be useful in future courses.

考试栏目和主题 Exam Sections & Topics	题库 Question Pool	提出的问题 Questions Presented	考试栏目和主题 Exam Sections & Topics	题库 Question Pool	提出的问题 Questions Presented
1 栏目 咖啡生豆 7% Section GREEN COFFEE 7%			子主题：烘焙过程中酸度变化 Sub-Topic: Acidity Changes during Roasting	6	3
1 主题：咖啡生豆的物理属性 Topic: Green Coffee Physical Attributes	Practical Exam		子主题：酸度随发展时间而变化 Sub-Topic: Acidity Changes Due to Development Time	3	1
2 主题：咖啡生豆化学 Topic: Chemistry of Green Coffee			4 主题：化学变化及其对苦味的影响 Topic: Chemical Changes and Their Effect on Bitterness	2	1
子主题：咖啡生豆的主要化学成分 Sub-Topic: Major Chemical Components of Green Coffee	3	1	5 主题：化学变化及其对芳香族的影响 Topic: Chemical Changes and Their Effect on Aromatics		
子主题：赭曲毒素（OTA）的原因和健康风险 Sub-Topic: Causes of Ochratoxins (OTA) and Health Risks	3	1	子主题：芳香族类别 Sub-Topic: Categories of Aromatics	2	1
2 栏目 咖啡烘焙中的热力学 20% Section THERMODYNAMICS IN COFFEE ROASTING 20%			子主题：芳香复杂性 Sub-Topic: Aromatic Complexity	2	1
1 主题：传播热量 Topic: Heat Transfer			6 主题：丙烯酰胺的形成 Topic: Acrylamide Formation	3	1
子主题：传热模式 Sub-Topic: Heat Transfer Modes	3	1	5 栏目 感官分析 4% Section SENSORY ANALYSIS 4%		

子主题：从烘焙炉到豆的热传递 Sub-Topic: Heat Transfer from Roaster to Bean	3	1	1 主题：评估发展时间调控 Topic: Evaluation of Development Time Modulations	3	1
2 主题：热扩散 Topic: Heat Diffusion			2 主题：质量控制方法 Topic: Quality Control Methodology	实践考试 Practical Exam	
子主题：热扩散基础 Sub-Topic: Basics of Heat Diffusion	2	1	6 Section WHOLESALE AND RETAIL BUSINESS 7%		
子主题：热量从外豆扩散到内豆 Sub-Topic: Heat Diffusion from Outer to Inner Bean	3	1	1 主题：客户偏好 Topic: Customer Preference	授课，不考 Taught, not tested	
子主题：水在热扩散中的作用 Sub-Topic: Water's Role in Heat Diffusion	3	1	2 主题：价格计算 Topic: Price Calculations		
3 主题：传热和扩散及其对烘焙瑕疵的影响 Topic: Heat Transfer and Diffusion and Their Effect on Roast Defects	2	1	子主题：烘焙咖啡的成本 Sub-Topic: Cost of Roasted Coffee	2	1
3 栏目 物理变化 14% Section PHYSICAL CHANGES 14%			子主题：利润率 Sub-Topic: Profit Margins	2	1
1 主题：膨胀 Topic: Expansion			Sub-Topic: Batch Size Requirements Related to Yearly Turnover Calculations	2	1
子主题：玻璃化转变 Sub-Topic: Glass Transition	3	1	7 栏目 烘焙厂管理 14% Section ROASTERY MANAGEMENT 14%		
子主题：内部压力 Sub-Topic: Internal Pressure	授课，不考 Taught, not tested		1 主题：烘焙生产设计与优化 Topic: Roastery Production Design and Optimization	2	1
子主题：孔隙率和结构退化 Sub-Topic: Porosity and Structural Degradation	2	1	2 主题：供应链管理 Topic: Supply Chain Management	授课，不考 Taught, not tested	
2 主题：萃取度和溶解度的变化 Topic: Changes in Extractability and Solubility			3 主题：监控烘焙过程 Topic: Monitoring Roasting Process		
子主题：烘焙度对可萃取性和溶解度的影响 Sub-Topic: Roast Degree's Effect on Extractability and Solubility	3	1	子主题：颜色测量的变化 Sub-Topic: Variations of Color Measurements	实践考试 Practical Exam	
子主题：烘焙速度对可萃取性和溶解度的影响 Sub-Topic: Roast Speed's Effect on Extractability and Solubility	3	1	子主题：烘焙记录软件 Sub-Topic: Roast Log Software	实践考试 Practical Exam	
4 栏目 化学变化 34% Section CHEMICAL CHANGES 34%			4 主题：拼配 Topic: Blending		
1 主题：化学变化和色值 Topic: Chemical Changes and Roast Color			子主题：拼配数量 Sub-Topic: Number of Blend Components	3	1
子主题：颜色变化的化学原因 Sub-Topic: Chemical Reasons for Color Change	4	1	子主题：前/后拼配 Sub-Topic: Pre / Post Blending	3	1

子主题：测量色值 Sub-Topic: Measuring Roast Color		实践考试 Practical Exam		5 主题：烘焙度对清洁和维护的影响 Topic: Roast Degree's Effect on Cleaning and Maintenance	2	1
2 主题：化学变化及其对溶解度的影响 Topic: Chemical Changes and Their Effect on Solubility		2	1	考试总题数 TOTAL QUESTIONS ON EXAM	76	30
3 主题：化学变化及其对酸度的影响 Topic: Chemical Changes and Their Effect on Acidity						

4. 带有相应在线笔试问题的课程 Course Curriculum with Corresponding Online Written Exam Questions

该课程的课程设置如下，分为栏目，主题和目标。在课程的某些栏目，创作者可能已经做了相对的修订，以创建更合乎逻辑，更适合层次的结构。任何修订均在 2。 *课程说明和更新*。 The course curriculum is set forth below and is divided into Sections, Topics and Objectives. In some areas of the curriculum, the creators may have revised the curriculum in order to create a more logical, level-appropriate structure. Any revisions are noted in 2. *Course Description and Updates*.

开发所有在线笔试题作为对特定目标的评估。这些问题已根据主题分组。主题内的所有问题均被视为主题“池”，也就是题库。从该库中，将随机选择一定数量的问题，并将其呈现给学习者。如果特定主题的目标不止一个，则有可能无法针对该主题的所有目标对学习者进行测试。这是由于该主题中问题的随机性。课程中还包括 AST 的注释，这些注释有助于说明内容以及如何实现目标。 All online written exam questions were developed as an assessment for a specific objective. These questions have been grouped according to topic. All questions within a topic are considered the topic “pool.” From this pool, a certain number of questions will be randomly selected and presented to the learner. If a particular topic has more than one objective, there is a possibility that the learner will not be tested on all objectives in the topic. This is due to the randomization of the questions from that topic. Also included in the curriculum are notes for the ASTs that help explain the content and how to achieve the objectives.

3.01 |栏目|咖啡生豆| Section | GREEN COFFEE

2 主题

主题 Topic	子主题 Subtopic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.01.01 咖啡生豆的 物理属性 Green Coffee Physical Attributes		1. 分析咖啡生豆的属性 (即水分读数和密度 读数) Analyze green coffee attributes (i.e. moisture reading and density reading)	建议学习者通过筛网查看豆子尺寸的整体一致性, 确定尺寸是否相对均匀。如果不一致, 建议您采用另一种烘焙方法 (慢速烘焙), 以确保烘焙出的咖啡豆大小均匀。 Recommend learners to look at the overall consistency of the screen size and see if the size is relatively uniform or not. If it is not consistent, recommend a separate roasting approach (slower roasting) to ensure roaster doesn't end up with some beans rare and some well-done. SCA 建议的水分含量为 8-12%。8-12% is the moisture content recommended by SCA. - 高于 12%, 存在发霉的风险 Lower than Higher than 12%, allows for a risk of mold - 低于 8%, 在烘焙过程中可能会损坏咖啡豆的组织 Lower than 8%, there is a risk of tissue damage to the coffee beans during roasting	通过实践考试 Tested on Practical Exam	Jansen, G., <i>Coffee roasting</i>, 8-11 Illy & Viani, <i>Espresso Coffee</i>, 135-140 Belitz, H.D., Grosch, W. & Schieberle, P., <i>Food</i>, 940

3.01.02 咖啡生豆化学 Chemistry of Green Coffee	3.01.02.01 咖啡生豆的主要化学成分 Major Chemical Components of Green Coffee	1. 了解咖啡生豆中化学成分的百分比 Know the percentages of chemical components found in green coffee		问题编码 Question ID 100009198220 哪种物质占阿拉比卡咖啡生豆总成分的 50%以上? Which substance represents more than 50% of the total components of green arabica coffee? 脂质 Lipids 蛋白 Proteins 多糖 Polysaccharides 糖氨基酸 Amino Acids 问题编码 Question ID 100009198223 在咖啡生豆中发现的脂质平均百分比是多少 What is the average percentage of lipids found in raw green coffee beans? 50% 8% 16% 25% 问题编码 Question ID 100009198229 将阿拉比卡咖啡生豆中的平均百分比与相应化合物匹配。 Match the average percentage found in green arabica coffee with the corresponding compound. 1) 8% 2) 50% 3) 15% 4) 10% a) 脂质 Lipids = 3 b) 蛋白质 Proteins = 4 c) 多糖 Polysaccharides = 2 d) 寡糖 Oligosaccharides = 1	Illy & Viani, <i>Espresso Coffee</i>, 148-160 Jansen, G., <i>Coffee roasting</i>, 11 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 274 - 280 Rao, S. <i>The Coffee Roaster's Companion</i>, 2-3 Belitz, H.D., Grosch, W. & Schieberle, P., <i>Food</i>, 940
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	3.01.02.02 赭麴毒素 (OTA) 的原因和健康风险 Causes of Ochratoxins (OTA) and Health Risks	1. 了解咖啡中 OTA 的主要载体以及与 OTA 接触相关的实际健康风险 Know the main vectors for OTA in coffee, and the actual health risks associated with OTA exposure	赭麴毒素是产地国于后製过程中由真菌所产生的健康风险因子。Ochratoxins are potential health risk factors developed by fungi during processing in the production countries. 在烘焙过程中，赭麴毒素的浓度会降低，但不会消除。 The concentration of Ochratoxins will be reduced, yet not eliminated, during the roasting process. 品质较好的咖啡生豆将含有较少的赭麴毒素。 The better grade of green coffee beans will contain less Ochratoxins. 目前，人们认为饮食中的 OTA 暴露量足够低，不会引起重大风险。 At the moment, dietary exposure to OTA is considered low enough to not cause significant risk.	问题编码 Question ID 100009198230 咖啡中的赭麴毒素该如何清除？ How can OTA be eliminated in coffee? 烘焙深一些 Roast the coffee dark. 烘焙浅一些 Roast the coffee light. <u>赭曲毒素无法藉由烘焙消除 OTA cannot be eliminated by roasting.</u> 将烘焙后的咖啡豆储存在单向透气阀的袋子裡 Store the coffee in one-way valve bag after roasting. 问题编码 Question ID 100009198231 导致咖啡生豆產生霉菌的主要媒介是什么？ What are the main vectors of green coffee that cause mold formation? 存放新鲜果實 Storage of fresh cherries 晒床晒干 Patio sun drying 受潮再润湿 Re-wetting 外壳储存污染 Husk storage contamination <u>以上所有 All of above</u> 问题编码 Question ID 100009198232 以下陈述是对还是错？赭麴毒素是产地国仅于后製过程中由真菌所产生的健康风险因子。 Is the following statement true or false? Ochratoxins are potential health risk factors developed by fungi exclusively during processing in production countries. <u>对 True</u> 错 False	Illy & Viani, <i>Espresso Coffee</i>, 163-164, 209, 368 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 68 Clarke, R. and Vitzhum, R. <i>Coffee Recent Developments</i>, 231 Van der Stegen, G., <i>Enhancement of coffee quality by mould prevention</i>, 245-249 Borém, <i>Post-Harvest Technology</i>, 217-231
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3.02 |栏目|咖啡烘焙中的热力学| Section | THERMODYNAMICS IN COFFEE ROASTING

3 个主题

主题 Topic	子主题 Sub-topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.02.01 传播热量 Heat Transfer	3.02.01.01 传热模式 Heat Transfer Modes	1. 在对流，传导和辐射方面分析和解释给定的烘焙技术 Analyze and explain a given roasting technology with regards to convection, conduction, and radiation	本节以来自中级的传导，对流和辐射方面的知识为基础。在专业层面，人们更加关注传热动力学。 This section builds on the knowledge from Intermediate in terms of conduction, convection and radiation. At the professional level, there is a stronger focus on the dynamics of heat transfer.	问题编码 Question ID 100009198233 什么是辐射热传递？ What is radiative heat transfer? 当热源通过空气直接传递到被加热物体时，能量转移。 The transfer of energy when a heat source passes through the air directly to the object being heated. <u>电磁波从一个物体到另一个物体的传输（例如，通过真空传播的辐射）。 The transfer of electromagnetic waves from one object to another (e.g., radiation travelling through a vacuum).</u> 气体中的分子振动从热源扩散到豆子，通过气流传输并辐射到豆子上。 The molecular vibration in gasses that diffuses from the heat source to the bean, transported by the airflow and radiated onto the beans. 以上都不是 None of the above 问题编码 Question ID 100009198234 在现代烘焙系统中，哪种类型的传热对咖啡的整体加热有最大的贡献？ Which is the type of heat transfer that has the largest contribution to the overall heating of the coffee in a modern roasting system? 传导 Conduction <u>对流 Convection</u> 辐射 Radiation	Clarke, R. and Vitzthum, R. <i>Coffee Recent Developments</i>, 90-100 Illy & Viani, <i>Espresso Coffee</i>, 187-190 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 256 Harold McGee, <i>Food & Cooking an Encyclopedia of Kitchen Science, History and Culture</i>, 780-782

				问题编码 Question ID 100009198235 以下陈述是对还是错？当热量从热空气直接传递到豆表面时，就会发生传导。 Is the following statement true or false? Conduction occurs when heat is transferred from the hot air directly to the bean surface. 对 True 错 False	
	3.02.01.02 从烘焙炉到豆的热传递 Heat Transfer from Roaster to Bean	1. 了解气温和热传递率之间的关系 Understand the relationship between air temperature and the rate of heat transfer	由于材料之间的热传递，传导热将从烘焙炉的加热元件扩散到烘焙炉的滚筒中，并且空气会通过系统。 As a consequence of heat transfer between materials, conductive heat will diffuse from the heating element of the roaster into the drum of the roaster and the air moving through the system. 然后，来自滚筒的热量和系统中的空气传递到豆子表面。然后，来自豆表面的热量扩散/传递到豆内部，并从一个豆的表面扩散到另一个表面。 The heat from the drum and the air in the system then transfers onto the bean surface. The heat from the bean's surface then diffuses/transfers into the bean's interior and from one bean's surface to the other.	问题编码 Question ID 100009198236 以下陈述是对还是错？在典型的生产烘焙机（滚筒）中，对流传递的热量比传导传递的热量更重要。Is the following statement true or false? In a typical production roaster (drum), convection transfers heat more significantly than conduction. 对 True 错 False 问题编码 Question ID 100009198237 以下哪个短语定义为烘焙过程中相对于咖啡豆的批处理量使用的热空气量？ Which phrase below is defined as the amount of hot air used in a roasting process in relation to the batch size of the coffee beans? 传热率 Rate of heat transfer 空豆比 (ABR) Air-to-bean ratio (ABR) 热能率 Rate of heat energy 烘焙速度率 Rate of roast speed	Jansen, G., <i>Coffee roasting</i>, 27-39 Illy & Viani, <i>Espresso Coffee</i>, 184- 187 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 256 Clarke, R. and Vitzhum, R. <i>Coffee Recent Developments</i>, 94-100 Roast Magazine, <i>The book of Roast</i>, 175

				问题编码 Question ID 100009198238 如果在烘焙过程中进/排气口的温度明显高于豆子的温度（假设空气温度高于豆子的温度），可以预期以下结果？ If the inlet/exhaust temperature is significantly greater than the bean temperature during roasting (assuming higher air temperature than bean temperature) which of the following outcomes can be expected? 豆子的温度升高速度较慢 A slower rate of temperature gain by the beans. 豆子获得更高的温度增加率 A higher rate of temperature gain by the beans. 不太可能发生尖端焦 The less likely tipping will occur. 温升速率将不受影响 The rate of temperature gain will not be affected.	
3.02.02 热扩散 Heat Diffusion	3.02.02.01 热扩散基础 Basics of Heat Diffusion	1. 了解，可视化并解释体内的热量“传导” Understand, visualize and explain ‘conduction’ of heat within a body	当存在温差时，分子振动会在材料之间以及内部内部扩散热量，热量会从较高的温度扩散到较低的温度。 Heat as molecular vibration diffuses between materials and internally within a material as soon as there is a temperature difference where heat can diffuse from higher temperature to lower temperature.	问题编码 Question ID 100009198239 以下哪个陈述最能说明从烘焙机到豆子的热传导 Which statement below best describes heat conduction from a roaster to the beans? 当通过与烘焙机硬表面的物理接触，热能从较热的物体传递到较冷的物体。 When, through physical contact with the hard surfaces of the roaster, heat energy passes from the hotter to the cooler object 当豆子堆成一堆并通过传导将它们的热能互相传递 When the beans lay in a pile, and through conduction transmit their heat energy to one another 辐射波从热的物体移动到冷的物体时 When waves of radiation move from the hot object to the cool object 空气将热能从热源传递到豆子时 When air carries heat energy from the heat-source to the beans	Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 256 Illy & Viani, <i>Espresso Coffee</i>, 183-184 Clarke, R. and Vitzhum, R. <i>Coffee Recent Developments</i>, 98

				问题编码 Question ID 100009198240 较高的预热如何导致烘焙开始时的速度增加 How does a high pre-heating cause a speed increase in the beginning of the roast? A) 箱内较热的空气温度将增加对流热能传递量 The hotter air temperature in the chamber will increase the amount of convective heat energy transfer. B) 较热的烘焙机将透过与金属接触，在烘焙开始时将更多的能量传递给咖啡豆。 Hotter roasters will have more energy to pass to the beans at the beginning of the roast, through contact with the hot metal. C) A 和 B A and B D) 较热的预热将导致热电偶错误地显示出较快的烘焙 The hotter pre-heating will cause the thermocouple to falsely show a faster roast.	
	3.02.02.02 热量从外豆扩散到内豆 Heat Diffusion from Outer to Inner Bean	1. 了解热能从豆表面向内的传递 Understand the transfer of heat energy from the surface of the bean inward	当热能接触到豆子时，它从外向内移动。咖啡从外面烘焙，这对于学习者理解以避免误解很重要。 When heat energy contacts the bean, it moves from the outside inward. Coffee roasts from the outside in, and this is important for learners to understand to avoid misconceptions.	问题编码 Question ID 100009198241 以下陈述是对还是错？至少烘焙过程的前 2/3 是吸热的，即豆吸收能量，热量从豆的外部移动到内部。 Is the following statement true or false? At least the first 2/3 of the roasting process is endothermic, i.e., beans absorb energy and heat moves from outside to inside the beans. 对 True 错 False	Clarke, R. and Vitzhum, R., <i>Coffee Recent Developments</i>, 98-99 Illy & Viani, <i>Espresso Coffee</i>, 183-184

			问题编码 Question ID 100009198242 咖啡烘焙本质上是对咖啡豆材料的加热，并且取决于以下哪个因素？ Coffee roasting is essentially a heating of the coffee bean material, and is dependent on which of the following factors? <u>豆子的内部导热性将热量从表面传递到中心。The internal thermal conductivity of the bean material to transfer heat from surface to center.</u> 红外辐射加热了豆中心。Infrared radiation heating up the bean center. 从烘焙开始就进行浸泡，以均匀加热豆子。 Performing the soak, to heat up the bean uniformly, from the beginning of the roast. 上述所有的 All of the above	
			问题编码 Question ID 100009198243 传导是加热内部咖啡豆结构的最重要因素，为什么？Is conduction the most important factor in heating up the internal bean structure and why? 不，对流更为重要，因为它可以更快地加热豆子。 No, convection is more important as it heats the bean faster. 不，辐射更重要，因为辐射将豆子加热到更深的内部。 No, radiation is more important as it heats the bean deeper inside. 是的，从滚筒表面到豆子表面的热量最有效。 Yes, heat from the drum surface to the bean surface is most effective. <u>是的，内部传导性确保热量分配。Yes, the internal conductivity ensures distribution of heat.</u>	

	3.02.02.03 水在热扩散中的作用 Water's Role in Heat Diffusion	1. 了解热量在豆子内扩散时的双重作用 Understand water's dual role when diffusing heat within the bean	水既可帮助在烘焙早期（100°C之前）传递热能，又可因汽化潜热而阻止热能。 Water both acts to help transmit heat energy early in the roast (before 100C) and then acts to resist heat energy due to the latent heat of vaporization.	问题编码 Question ID 100009198244 在烘焙过程中，水如何在豆子的物理变化中起重要作用？ How does water play an important role in the bean's physical changes during the roast? 由于水分的蒸发，它会在干燥阶段导致压力升高。It causes pressure build-up in the drying phase due to the water evaporation. 它是热量从豆子的表面扩散到豆子中心的重要介质，因此也是烘焙初期降低豆子颜色渐层的重要角色。It is an important medium for the heat to diffuse in from the surface to the center of the bean and thereby also an important means for decreasing the color gradient in the bean at the beginning of the roast. 由于水具有大量吸热性的特质，它是在烘焙开始时保持温度不升高的重要物质。It is the substance that keeps the temperature from rising in the beginning of the roast due to the heavy endothermic nature of water. <u>以上全部 All of the above</u> 问题编码 Question ID 100009198245 特定咖啡温度曲线所需的热能取决于许多因素。哪一个最相关？ The thermal energy required for a specific coffee temperature profile depends on many factors. Which one is the most relevant? 咖啡豆来源 Coffee bean origin 咖啡豆品种 Coffee bean variety 干咖啡物质 Dry coffee substance <u>生豆水分 Water content of green bean</u>	Jansen, G., <i>Coffee roasting</i>, 20-21
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				问题编码 Question ID 100009198246 以下陈述是对还是错？一爆之前需要能量，以促进豆子中水分的蒸发。没有水分，热量会直接影响豆子原料，从而导致温度升高更快。 Is the following statement true or false? Energy is needed before first crack to facilitate the evaporation of moisture in the bean. Without moisture, the heat can directly impact the bean material, causing the temperature to increase faster. 对 True 错 False	
3.02.03 传热和扩散及其对烘焙瑕疵的影响 Heat Transfer and Diffusion and Their Effect on Roast Defects		1. 了解物理烘焙瑕疵的产生方式以及如何避免造成这些瑕疵 Understand how physical roast defects are generated and how to avoid creating them	传导热传递太快会引起焦化问题，而对流可能会使热传递（基于豆子的密度）太快（导致发烫）。 Too rapid of conductive heat transfer causes the issue of scorching, and too rapid of heat transfer (based on the density of the bean), likely from convection, causes tipping.	问题编码 Question ID 100009198247 如何检测外表烘焙瑕疵，例如熟豆表面上有烧焦點？ How are external roasting defects like scorching on roasted coffee detected? 豆子表面有黑点 Black spots on the surface of the bean 密度高的 High density of the bean 破碎豆子 Broken beans 以上所有 All of the above	
				问题编码 Question ID 100009198248 烘焙时会导致瑕疵“顶端焦”的原因是什么？ What could cause the defect "tipping" when roasting? 滚筒表面太冷 Too cold of a drum surface 滚筒表面太热 Too hot of a drum surface 烘焙太快 Too fast of a roast 烘焙太慢 Too slow of a roast	

3.03 | Section | PHYSICAL CHANGES

3 个主题

主题 Topic	子主题 Sub-topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.03.01 膨胀 Expansion	3.03.01.01 玻璃化转变 Glass Transition	1. 解释咖啡的玻璃化转变及其在烘焙过程中的意义 Explain the glass transitions of coffee and their significance in the roasting process	室温下的生豆在结构外观上类似于玻璃 Green beans at room temperature are 'glass like' in their structural appearance. 当达到玻璃化转变温度时，生豆将变成“皮革状”Polymeric material from the green beans will become 'leathery' when the glass transition temperature is reached. 因此，生豆在烘焙过程中呈皮革状，但到达冷却托盘后不久又变成玻璃状。	问题编码 Question ID 100009198249 咖啡烘焙过程中玻璃化转变的意义是什么？ What is the significance of the glass transition during coffee roasting? 玻璃化转变不会对烘焙过程中的咖啡豆产生重大影响。 Glass transition does not significantly affect the coffee beans during roasting. 玻璃化转变直接导致风味的产生。 Glass transition is directly responsible for flavor generation. <u>玻璃化转变使压力的积累和咖啡豆中化学反应的途径成为可能。</u> <u>Glass transition makes the build-up of pressure, and pathways of chemical reactions in the coffee bean possible.</u> 玻璃化转变解释了为什么银皮在一爆就从咖啡豆中脱落了。 Glass transition explains why the chaff comes off the coffee bean post first crack.	Jansen, G., <i>Coffee roasting</i>, 32-34 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 247

		这是二阶相变（AST 可以使用水作为一个粗略的例子）。 Therefore, green beans are leathery during the roasting process but become glass like again soon after reaching the cooling tray. This is a second order phase transition (the AST could use water as a rough example). 要点：玻璃化转变使咖啡豆在烘焙过程中随压力增加而溶胀（与破碎相反）， Essential Point: Glass transition allows the coffee bean to swell along with pressure increases during roasting (as opposed to breaking) and allows for volume increase. 玻璃化转变温度根据水分含量而变化。从玻璃到皮革的过渡发生在结束时间附近，通常在冷却过程中回到玻璃态。 Temperature for glass transition varies based on moisture content. Early transition from glass to leather happens near turnaround, and typically crosses back to glassy during cooling.	问题编码 Question ID 100009198250 为了使咖啡豆在烘焙过程中發生膨胀，必须有什么？ What is necessary in order for the swelling of the bean material to occur during roasting? <u>水的蒸发和咖啡豆從玻璃态转变为皮革状的过程，会使咖啡豆在烘焙过程中膨胀。The evaporation of water and transition of the bean material from a glassy to leathery state allows the bean to swell during roasting.</u> 对流热将咖啡豆中的化合物点燃，形成了新的更大的化合物，它们占据了更多的空间，从而增加了咖啡豆的大小。 The ignition of the compounds in the bean material by convective heat forms new and larger compounds that take up more space, thereby increasing the size of the bean. 由于存在蔗糖和随后的烘焙过程中的焦糖化作用，咖啡豆会膨胀。 Coffee beans swell due to the presence of sucrose and the subsequent caramelization during roasting. 烘焙机中较高的气流有助于使咖啡豆在烘焙过程中膨胀并展开。 A higher airflow from the roaster helps to cause the beans to swell and expand during the roast.	
			问题编码 Question ID 100009198251 选择最能完成以下句子的答案。玻璃化转变阶段使咖啡豆结构 ____。 Select the answer that best completes the following sentence. The glass transition phase allows the bean structure to ____. A) 在烘焙过程中暂时变得更灵活，从而能够膨胀。 temporarily become more flexible during roasting and thereby able to expand B) 烘焙后变硬变脆，使结构更容易断裂。 become hard and brittle after roasting which makes the structure easier to break. <u>C) A 和 B both A and B</u>	

	3.03.01.02 内部压力 Internal Pressure	1. 解释水蒸气和其他烘焙气体在烘焙过程中压力升高方面的作用 Explain the role of water vapor and other roast gases with regards to pressure increase during roasting	加热时，首先水（生豆的水分 8-12%）将变成蒸气。后来，梅纳反应，Strecker 降解，有机酸的焦糖化和脱羧作用将产生有机烘焙气，例如二氧化碳（CO ₂ ）和其他小的有机气 First water (8-12% moisture of the green bean) will turn into vapor when heated. Later the Maillard reaction, Strecker degradation, caramelization and decarboxylation of organic acids will create organic roast gasses like carbon dioxide (CO ₂) and other small organic gasses.	授课，不考。 Taught, not tested.	Clarke, R. and Vitzhum, R. <i>Coffee Recent Developments</i> , 94-95 Rao, S., <i>The Coffee Roaster's Companion</i> , 12
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	3.03.01.03 多孔性和结构分解 Porosity and Structural Degradation	1. 解释咖啡豆在膨胀和内部压力增加后，纤维素结构是如何变化的。 Explain how the cellular structure of the coffee bean changes due to expansion and internal pressure	随着咖啡豆的膨胀，现有的细胞孔也会膨胀，从而在咖啡豆内形成更大的开口。与此相伴的是，烘焙过程中产生的内部压力以及豆细胞结构破损，使小裂缝形成，从而在细胞基质中留下了更多的开口。 As the coffee bean expands, the existing cellular pores also expand, creating larger openings within the bean. 具有发展中的气体（蒸气，CO₂ 和有机气体）的“皮革状”将使咖啡膨胀，致密结构变成具有许多小气穴和通道的海绵状结构。 Along with this, the internal pressure created during roasting and the now compromised integrity of the bean's cellular structure allow small fissures to form, leaving even more openings in the cellular matrix. The 'leathery' material with developing gasses (vapor, CO₂ and organic gases) will expand and the dense material will become a spongy structure with many small gas pockets and channels. 油可以通过这些通道传播，因此，如果不仔细调整好烘焙曲线控制烘焙程度的话，机油会迁移到表面并在空气中氧化。	问题编码 Question ID 100009198252 “发汗咖啡”是当烘焙后咖啡豆表面上出现细微的脂肪滴时的一种现象。是什么原因造成的？选择两 (2) 个正确答案。 'Sweating coffee' is a phenomenon that occurs when fine drops of fat are present on the surface of the bean after roasting. What can cause this? Select the two (2) correct answers. 快速浅烘 Fast light roasting (8:00 - 10:00) 深烘焙（比二爆的最终温度更高） Dark roasting (Higher end temperature than second crack) 慢速浅烘 Slow light roasting (13:00 - 15:00) 在过高的温度下干燥咖啡生豆 Drying the green coffee at too high temperature	Illy & Viani, <i>Espresso Coffee</i>, 182-184 Clarke, R. and Vitzthum, R., <i>Coffee Recent Developments</i>, page 95 Jansen, G., <i>Coffee roasting</i>, 45 Rao, S., <i>The Coffee Roaster's Companion</i>, 12-13
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			Oils can travel through these channels so if a careful roast profile and roast degree are not used, oils can then migrate to the surface and become rancid when oxygen from the air reaches the oils. 注意：请记住，深烘焙是一种风格偏好，也是一种客户偏好。 Note: Remember that roasting dark is a stylistic preference and a customer preference.	问题编码 Question ID 100009198253 讨论咖啡豆结构时，“皮革状”是指什么？ What does “leathery” refer to when discussing coffee bean structure? 豆变脆变多孔的状态，使研磨更加均匀。 A state where the bean becomes brittle and porous which makes grinding more uniform. 咖啡豆变得更柔韧的状态允许结构的重塑 A state where the coffee bean becomes more flexible allowing for remodeling of the structure. 豆子在表面有皱纹的地方变得光滑的状态 A state where the bean becomes glossy on the surface with wrinkled areas.	
3.03.02 萃取度和溶解度的变化 Changes in Extractability and Solubility	3.03.02.01 烘焙度对可萃取性和溶解度的影响 Roast Degree's Effect on Extractability and Solubility	1. 了解烘焙程度对可溶性化合物萃取的影响 Understand the impact of roast degree on extraction of soluble compounds	烘焙越深，豆子的材料就会变得越多孔和越脆，就会出现较大的内部通道，水可以通过这些通道流动并收集可溶性分子。豆子的材料的多孔性和脆性越小，通道将越小，从而导致水萃取较稠密材料中的可溶性分子的能力降低。	问题编码 Question ID 100009198254 较深的烘焙咖啡导致冲泡咖啡中可溶物浓度增加的原因是什么？ What is the reason that a darker roasted coffee leads to an increased solubles concentration extraction in brewed coffee? A) 较深的烘焙较脆，在研磨时产生更多细粉。 A darker roast is more brittle, yielding more fines upon grinding. B) 较深的烘焙，内部的“通道”较大，这使水更容易进入研磨中心并萃取更多可溶化合物。 The darker the roast, the larger the internal 'channels' which makes it easier for water to enter the center of the grounds and extract more soluble compounds. <u>C) A 和 B A and B</u>	Jansen, G., <i>Coffee roasting</i>, 22-27 Alstrup, J., SCA News, 25 Magazine, <i>Modulating Perception</i>

			较深的烘焙较脆，在研磨时会产生更多的咖啡粉，而较深的烘焙可溶物则更快地溶解（萃取）。 The darker the roast, the more porous and brittle the bean material will be, with larger internal channels occurring through which the water can run and harvest the soluble molecules. The less porous and brittle the bean material, the smaller the channels will be, causing the water to be less able to extract the soluble molecules in the denser material. A darker roast is more brittle, yielding more ground coffee fines upon grinding and darker roasted soluble matter dissolves (extracts) quicker.	问题编码 Question ID 100009198255 烘焙度如何影响可萃取率和溶解度?How does roast degree impact extractability and solubility? 从较深的烘焙咖啡中只能萃取苦味化合物。Only bitter compounds will be extracted from darker roasted coffees. 较深的烘焙较脆，在研磨时会产生更多细粉，从而提高萃取率。 A darker roast is more brittle, yielding more fines upon grinding, leading to higher yield extraction. 较浅的烘焙具有较大的内部通道，可促进水萃取更多的可溶性化合物。A lighter roast has larger internal channels that facilitate water to extract more soluble compounds. 烘焙程度不影响萃取率，仅影响溶解性。Roast degree does not impact extractability, only solubility.	
				问题编码 Question ID 100009198256 深烘焙咖啡具有更大的体积，更多和更大的内部孔隙，更高的孔隙率和更大的总体表面积。这导致以下哪项结果？ Dark roasted coffees have been shown to have higher volume, more and larger internal pores, higher porosity and larger surface area overall. This results in which of the following outcomes? 更愉悦的风味和口感 More pleasant flavors and body 更高的萃取率 Higher extractability in brewing 可保存较久的新鲜度 Fresher coffee for a longer time period 更高的咖啡品质 Higher cup quality	
	3.03.02.02 烘焙速度对可萃取性和溶解度的影响 Roast Speed's Effect on Extractability and Solubility	1. 了解烘焙速度对可溶性化合物萃取的影响 Understand the impact of roast speed on extraction of soluble compounds	由于内部压力以及细胞结构软化和降解，更快地烘焙的咖啡往往具有更高的总体积增加。这种增加的体积使水更容易渗透到研	问题编码 Question ID 100009198257 以下陈述是对还是错？快速烘焙的咖啡往往更易溶解，因为豆子的细胞保持紧密得多。 Is the following statement true or false? Quickly roasted coffee tends to be more soluble because the bean's cells stay much more closed. 对 True 错 False	Alstrup, J., SCA News, 25 Magazine, <i>Modulating Perception</i> Clarke, R. and Vitzhum, R., <i>Coffee Recent Developments</i>, 92-95

			磨物中，从而允许更多的萃取。 Coffee roasted more quickly tends to have a higher overall volume increase due to internal pressure and the cellular structure softening and degrading. This increased volume allows water to infiltrate the grinds more readily and thus allow for more extraction.	问题编码 Question ID 100009198258 从烘焙机到豆子的快速热传递将导致哪项结果？ Fast heat transfer from roaster to bean will reliably cause which outcome? 低萃取率 Low yield extractability 豆细胞壁中较大的微孔 Larger micropores in the bean cell walls 豆体积较大 Greater bean volume 油向豆表面的迁移 Oil migration to bean surface	Illy & Viani, <i>Espresso Coffee</i>, 190
				问题编码 Question ID 100009198259 如何改善咖啡粉中可溶物的萃取率？ How can the extractability of solubles from the coffee grounds be improved? 长时间浅烘 Long roast time with a light end color 快速烘焙 Fast roast speed 烘焙初期很温柔 Very gentle beginning of the roast 更大的对流 Higher airflow	

3.04|栏目|化学变化 | Section | CHEMICAL CHANGES

6 主题

主题 Topic	子主题 Sub-topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.04.01 化学变化和色 值 Chemical Changes and Roast Color	3.04.01.01 颜色改 变的化学原因 Chemical Reasons for Color Change	1. 解释各种糖褐变 反应及其对颜色的 影响 Explain the various sugar browning reactions and their effect on color		问题编码 Question ID 100009198260 选择最适合该语句的术语 Select the term that best matches the statement. _____A_____在氧气和酶的作用下发生，导致细胞中的颗粒变成褐色产品，使咖啡豆呈现典型的棕色。 _____A_____ occurs under the influence of oxygen and enzymes which causes particles in the cells to change into browning products and give the coffee beans a typical brown color. _____B_____在热的影响下发生，形成带有棕色色素的化合物。_____B_____ occurs under the influence of heat, forming compounds with brown pigmentation. <u>A =酶促褐变 Enzymatic browning</u> <u>B =非酶褐变 Non-enzymatic browning</u>	Illy & Viani, <i>Espresso Coffee</i>, 204-208 Jansen, G., <i>Coffee roasting</i>, 48, 64 Harold McGee, <i>Food & Cooking an Encyclopedia of Kitchen Science, History and Culture</i>, 777-780 Roast Magazine, <i>The book of Roast</i>, 161-165 Clarke, R. and Vitzhum, R., <i>Coffee Recent Developments</i>, 13-14
				问题编码 Question ID 100009198261 以下哪个说法是正确的 Which statement below is correct? <u>焦糖化是加热糖分子引起的脱水褐变反应 Caramelization is a dehydration browning reaction caused by heating sugar molecules.</u> 梅纳反应是加热糖分子引起的脱水褐变反应 Maillard reaction is a dehydration browning reaction caused by heating sugar molecules. 焦糖化是加热糖分子引起的再水合褐变反应 Caramelization is a rehydration browning reaction caused by heating sugar molecules. 梅纳反应是由于加热氨基酸引起的再水合褐变反应 Maillard reaction is a rehydration browning reaction caused by heating amino acids.	

			问题编码 Question ID 100009198262 烘焙咖啡时，会发生哪两 (2) 次褐变反应？ Which two (2) browning reactions take place when roasting coffee? 焦糖化 Caramelization 梅纳反应 Maillard reaction 量热法 Calorimetry 水解 Hydrolysis	
			问题编码 Question ID 100009198263 在咖啡烘焙过程中，必须存在哪些两 (2) 种必要的催化剂才能发生褐变反应？ Which two (2) essential catalysts must be present for browning reactions to occur in coffee roasting? 氨基酸 Amino Acids 热 Heat 糖 Sugar 酶 Enzymes	

	3.04.01.02 测量色值 Measuring Roast Color	1. 正确使用烘焙颜色测量工具 Use roast color measurement tools correctly	具体来说，学习者应该能够使用色值测量设备并解释以下结果： Specifically, the learner should be able to handle roast color measurement equipment and interpret the results for: 1. 用于烘焙策略设计 use in general profile product development and 2. 和把它用作分析手段，以用于在生产中评估出现不同情形的烘焙 as an analysis of variation tolerance for a product at any given time during normal production processes.	在实操考试中考核 Tested on Practical Exam.	
3.04.02 化学变化及其对溶解度的影响 Chemical Changes and Their Effect on Solubility		1. 说明改变烘焙度时发生溶解度变化的化学原因 Explain the chemical reasons for changes in solubility that occur when the roast degree is changed	碳水化合物的水解使大多数糖更易溶（极性）且更小，因此它们更容易穿过咖啡水基质。 Hydrolysis of carbohydrates makes large polysaccharides more soluble (polar) and smaller, so they move more easily through the coffee-water matrix. The di- and monosaccharides are cleaved into acids and other smaller polar compounds which allows them to move more quickly in the coffee water matrix.	问题编码 Question ID 100009198264 纤维素在烘焙过程中是否参与梅纳反应？ Does cellulose participate in the Maillard reaction during roasting? 是的，因为纤维素是多糖。 Yes, because cellulose is a polysaccharide 否，因为在烘焙过程的任何阶段都不会降解纤维素。 No, because cellulose is not degraded at any stage of the roast process 是的，因为水解会将纤维素分子裂解为寡糖和单糖，然后供入梅纳反应中。 Yes, because hydrolysis cleaves the cellulose molecule into oligo- and monosaccharides that then feeds into the Maillard reaction 以上都不是 None of the above	Belitz, H., Grosch, W. and Schieberle, P., n.d. <i>Food Chemistry</i>, 270 Damodaran, S., Parkin, K. and Fennema, O., <i>Fennema's Food Chemistry</i>, 109-110

				问题编码 Question ID 100009198265 选择最能使句子完整的短语。在烘焙过程中，水解在 _____ 中起着重要作用。 Select the phrase that best completes the sentence. During roasting, hydrolysis plays an important role in _____. 增加碳水化合物的溶解度 increasing the solubility of carbohydrates 为梅纳反应提供反应物，从而增加咖啡的溶解度 providing reactants for the Maillard reaction thus increasing the solubility of the coffee 产生糖分，从而增加咖啡的甜味 creating sugars and thereby adding sweetness to the coffee brew	
3.04.03 化学变化及其对酸度的影响 Chemical Changes and Their Effect on Acidity	3.04.03.01 烘焙过程中酸度变化 Acidity Changes during Roasting	1. 列出在烘焙过程中咖啡生豆中存在的哪些酸发生变化以及它们如何变化 List which acids, present in green coffee, change during the roasting process and how they change	乙酸，乳酸，乙醇酸，甲酸，奎宁酸，咖啡酸，磷酸和琥珀酸在烘焙过程中都会减少。 Acetic, lactic, glycolic, formic, quinic, caffeic, phosphoric, and succinic acids all decrease during roasting.	问题编码 Question ID 100009198266 咖啡生豆中存在哪些酸，并且在烘焙过程中会减少？ Which acids exist in green coffee and decrease during roasting? 柠檬酸，琥珀酸，甲酸和酒石酸 Citric, succinic, formic and tartaric acids 乙酸，甲酸，乳酸和奎尼酸 Acetic, formic, lactic and quinic acids 乳酸，磷酸，苹果酸和酒石酸 Lactic, phosphoric, malic and tartaric acids 绿原酸，柠檬酸，酒石酸和苹果酸 Chlorogenic, citric, tartaric and malic acids 问题编码 Question ID 100009198267 通常，咖啡生豆中存在的酸在烘焙过程中会部分分解或转化。绿原酸的降解可以形成哪些两 (2) 种酸？ Generally, acids present in green coffee beans are partially decomposed or transformed during the roasting process. The degradation of chlorogenic acids can form which two (2) acids? 柠檬酸 Citric 酒石酸 Tartaric 奎宁酸 Quinic 咖啡酸 Caffeic	Clarke, R. and Vitzthum, R., <i>Coffee Recent Developments</i>, 18-30, 59 Illy & Viani, <i>Espresso Coffee</i>, 194-197

		2. 列出在烘焙过程中产生/增加的酸。 List which acids are created/increased during roasting.		问题编码 Question ID 100009198268 是什么原因导致奎宁酸和咖啡酸的形成？ What causes the formation of quinic and caffeic acid? 氨基酸脱羧 Decarboxylation of amino acids <u>绿原酸的分解</u> <u>Breakdown of chlorogenic acid</u> 咖啡豆中糖的发酵 Fermentation of sugars in the coffee bean 储存中氧化 Oxidation during storage 问题编码 Question ID 100009198269 烘焙过程中在烘焙过程后期出现的哪些酸尚未存在于咖啡生豆中？ Which acids, occurring later in the roasting process, are not already present in the green coffee prior to roasting? <u>乙酸，乳酸，甲酸和乙醇酸</u> Acetic, lactic, formic, and glycolic acids 磷酸、柠檬酸，苹果酸和奎宁酸 Phosphoric, citric, malic, and quinic acids 绿原酸，酒石酸，甲酸和乙酸 Chlorogenic, tartaric, formic, and acetic acids 柠檬酸，乳酸，酒石酸和苹果酸 Citric, lactic, tartaric, and malic acids 问题编码 Question ID 100009198270 烘焙过程中哪些酸会降解 Which acids are degraded during roasting? 乙酸，乳酸和柠檬酸 Acetic, lactic and citric acids <u>绿原酸，柠檬酸和苹果酸</u> Chlorogenic, citric and malic acids 绿原酸，酒石酸和乙酸 Chlorogenic, tartaric and acetic acids 酒石酸，乳酸和柠檬酸 Tartaric, lactic and citric acids
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				问题编码 Question ID 100009198271 烘焙过程中蔗糖的降解会产生哪种酸？ Sucrose degradation during the roasting process creates which acid? 苹果 Malic 醋酸 Acetic 绿原酸 Chlorogenic 咖啡酸 Caffeic	
	3.04.03.02 酸度随发展时间而变化 Acidity Changes Due to Development Time	1. 说明更长的发展时间往往会降低总酸度的印象 Explain how longer development times tend to yield lower impressions of overall acidity	一些酸性化合物是碳水化合物的衍生物（乙酸，甲酸，乳酸，乙醇酸等），在烘焙过程中很早就形成，后来随着咖啡在一爆时和一爆后变黑而降解。 Some acidic compounds are derivatives of carbohydrates (acetic, formic, lactic, glycolic, etc.) and are developed very early in the roasting process and later degraded as the coffee becomes darker around and after 1st crack.	问题编码 Question ID 100009198272 较长的发展时间会导致酸度降低。原因之一是绿原酸分解成两 (2) 种酸？ Longer development time can lead to lower acidity perception. One of the reasons for this is the breakdown of chlorogenic acid into which two (2) acids? 柠檬酸和乙酸 Citric and acetic acids 乳酸和甲酸 Lactic and formic acids 乙醇酸和奎宁酸 Glycolic and quinic acids 奎宁和咖啡酸 Quinic and caffeic acids	
				问题编码 Question ID 100009198273 较长的发展时间会导致哪些酸降解？ Longer development time leads to the degradation of which acids? 柠檬酸 Citric acid 醋酸 Acetic acid 绿原酸 Chlorogenic acid 上述所有的 All of the above	

				问题编码 Question ID 100009198274 从理论上讲，为什么更长的发展时间会导致咖啡的酸度降低？ Why would longer development time theoretically lead to a lower perception of acidity in the coffee? 苦味和果味的丧失增加 There is an increase in bitterness and loss of fruity notes. 有机酸含量增加 There is an increase in levels of organic acids 有机酸含量降低 There is a decrease in levels of organic acids.	
3.04.04 化学变化及其对苦味的影响 Chemical Changes and Their Effect on Bitterness		1. 说明烘焙过程中造成苦味的原因 Explain what causes bitterness during the roasting process	苦味主要归因于梅纳反应的产物，绿原酸的分解和焦糖化的产物。 Bitterness is primarily due to the products of the Maillard reaction, the breakdown of chlorogenic acids and the products of caramelization. 烘焙得越深，苦味就越大 The darker the roast the higher the bitterness.	问题编码 Question ID 100009198275 烘焙过程中发生的两 (2) 种苦味是什么？ What are the two (2) sources of perceived bitterness that occur during the roasting process? 烘焙过程中咖啡因的生成 Caffeine development during roasting 梅纳反应和焦糖化反应的产物 Products of the Maillard and caramelization reactions 绿原酸分解的产物 Products of the breakdown of chlorogenic acids 烘焙过程中的气流过多 Too much airflow during roasting	Clarke, R. and Vitzhum, R., <i>Coffee Recent Developments</i>, 53 Jansen, G., <i>Coffee roasting</i>, 48 Roast Magazine, <i>The book of Roast</i>, 209-215
				问题编码 Question ID 100009198276 以下陈述是对还是错？咖啡中发现的大多数苦味化合物是在梅纳反应期间产生的。 Is the following statement true or false? The majority of the bitter-tasting compounds found in coffee arise during the Maillard reaction. 对 True 错 False	

3.04.05 化学变化及其 对芳烃的影响 Chemical Changes and Their Effect on Aromatics	3.04.05.01 芳香族 的类别 Categories of Aromatics	1. 说明烘焙咖啡中 存在的 6 类挥发性 芳香族化合物 State the 6 categories of volatile aromatic compounds present in roasted coffee	在一般情况下，咖啡的味 道会随着烘焙程度而变 化；但是在更细微的程度 上，即使相同的烘焙色度 也可能表现出许多不同的 感官特性，具体取决于所 应用的特定温度曲线。 On a very general level, the flavor of coffee changes with roast degree; but on a more subtle level even the same roast color degree could express many different sensory properties depending on the specific temperature profile applied. 尽管烘焙咖啡中存在数千 种挥发物，但只有 28 种 芳香物质可以大致近似 “咖啡香气”。 While thousands of volatiles are present in roasted coffee only 28 aromatic substances can largely approximate 'coffee aroma' . 烘焙咖啡中存在以下类别 的挥发性芳香族化合物 The following categories of volatile aromatic compounds are present in roasted coffee: -土味 Earthy -果味 Fruity -甜/焦糖 Sweet / Caramel -酚醛/烟熏 Phenolic / Smoky -硫/烘烤 Sulphurous / Roasty -辛辣 Spicy	问题编码 Question ID 100009198277 烘焙咖啡中存在哪些种类的挥发性芳香族化合物？ Which categories of volatile aromatic compounds are present in roasted coffee? 坚果/可可，香料，酸/发酵，果味，硫/烘烤，其他 Nutty/cocoa, spices, sour/fermented, fruity, sulphurous/roasted, other <u>甜/焦糖，土味，硫/烘烤，烟熏/酚醛，果味，辛辣</u> <u>Sweet/caramel-like, earthy, sulphurous/roasted,</u> <u>smoky/phenolic, fruity, spicy</u> 花香，果味，甜/焦糖，辛辣，其他，土味 Floral, fruity, sweet/caramel-like, spicy, other, earthy 绿色/蔬菜，坚果/可可，酸/发酵，烟熏/酚醛，花香， 果味 Green/vegetative, nutty/cocoa, sour/fermented, smoky/phenolic, floral, fruity	Clarke, R. and Vitzhum, R., <i>Coffee Recent Developments</i>, 76 Table 3.6 Roast Magazine, <i>The book of Roast</i>, 199 - 201 Illy & Viani, <i>Espresso Coffee</i>, 197-201
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			与咖啡香气相关的最重要的化合物是：甜/焦糖。 The most significant category of compounds associated with coffee aroma is: sweet/caramel. (Reference: The book of Roast 199-201)	问题编码 Question ID 100009198278 什么种类的芳香化合物最常与咖啡香气有关？ What category of aromatic compounds is the most often associated with coffee aroma? 土味 Earthy 果味的 Fruity 甜/焦糖 <u>Sweet / Caramel</u> 酚醛/烟熏 Phenolic / Smoky 含硫/烘焙 Sulphurous / Roasty 辛辣 Spicy	
	3.04.05.02 芳香族的复杂性 Aromatic Complexity	1.知道咖啡的香气可能仅由少于 50 种的挥发性香气组成 Know that coffee aroma is likely only made up of less than 50 volatile aromatics	注意：与芳香剂相比，咖啡所含的数千种不同挥发性化合物的芳香性要低。 Note: Coffee is aromatically less complex than the thousands of different volatile compounds it contains.	问题编码 Question ID 100009198279 有多少主要的香气成分使咖啡闻起来像咖啡 How many key aroma compounds are involved in making coffee smell like coffee? 10-15 800-1000 40-60 28-30	
				问题编码 Question ID 100009198280 虽然烘焙咖啡中存在数千种挥发物，但多少种近似的气味（芳香物质）是造成“咖啡香气”的主要原因？ While thousands of volatiles are present in roasted coffee, how many approximate odorants (aromatic substances) are largely responsible for 'coffee aroma'? 20 28 13 800	

3.04.06 丙烯酰胺的形成 Acrylamide Formation		1. 描述烘焙过程中丙烯酰胺的形成和降解方式 Describe how acrylamide is formed and degraded during roasting	丙烯酰胺是烘焙过程中产生的另一种癌症危险因素，但随着时间的推移也会降解。因此，慢速烘焙咖啡几乎不含丙烯酰胺，而快速烘焙咖啡则含有相当多量的丙烯酰胺。 Acrylamide is another cancer risk factor developed during the roasting process, yet also degraded by time. So slow roasted coffee contains almost no acrylamide, whereas flash roasted coffee contains a considerable amount. 请记住要告知学习者，与我们经常食用的其他熟食或油炸食品相比，咖啡中的丙烯酰胺含量相对较小。 Remember to inform learners that the amount of acrylamide in coffee is relatively insignificant when compared to other cooked or fried foods that we regularly consume.	问题编码 Question ID 100009198281 比较不同烘焙曲线中的丙烯酰胺含量，可以预期哪一个曲线显示出最低的丙烯酰胺含量？ Comparing acrylamide levels in different roast profiles, which profile would be expected to show the lowest level of acrylamide? <u>慢速烘焙咖啡 Slow roasted coffees</u> 快速烘焙咖啡 Fast roasted coffees 慢速和快速烘焙咖啡 Slow and fast roasted coffees 烘焙曲线未显示丙烯酰胺数值 Roast profiles do not show acrylamide level 问题编码 Question ID 100009198282 选择有关丙烯酰胺的最正确陈述。 Select the most correct statement about acrylamide. <u>丙烯酰胺是梅纳反应的副产物 Acrylamide is a by-product of Maillard reactions.</u> 丙烯酰胺的热敏感性较低 Acrylamide has low heat sensitivity. 快烘中丙烯酰胺含量低 Flash roasts present low levels of acrylamide. 丙烯酰胺不构成任何健康风险 Acrylamide does not represent any health risk. 问题编码 Question ID 100009198283 以下陈述是对还是错？ 丙烯酰胺的形成发生在烘焙过程的后期。 Is the following statement true or false? Acrylamide formation occurs during the latter stages of the roasting process. 对 True 错 <u>False</u>	Roast Magazine, <i>The book of Roast</i>, 187-189 Guenther, H. et al., Food Additives and Contaminants, <i>Acrylamide in coffee: Review of progress in analysis, formation and level reduction</i>, 24, 60–70
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3.05 | 栏目|感官分析| Section | SENSORY ANALYSIS

2 主题

主题 Topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.05.01 评估发展时间 调控 Evaluation of Development Time Modulations	1. 识别由于发展时间调控而引起的风味差异 Identify flavor differences due to development times modulations	发展时间调控的示例：快速，参考和烘焙烘焙。 Example of development time modulations: fast, reference and baked roasting.	问题编码 Question ID 100009198284 如果在一爆和二爆之间的发展时间过长，那么烘焙咖啡的味道会怎样？ What happens to the flavor of a roasted coffee when the development time, in between the 1st and 2nd crack, is too long? 会尝起来平乏和焙烤味 It will taste dull and roast-like. 变甜 It will taste sweeter. 变果味 It will taste fruity. 变酸 It will taste more acidic. 问题编码 Question ID 100009198285 发展不足咖啡的特点是什么？What are characteristics of less-developed coffees? 开始杯测时杯测碗表面的浮起的咖啡渣可能会崩解。A cupping crust likely to disintegrate. 烘焙过程中焦糖化糖产物降低或二氧化碳气体的产生减少。 Reduced caramelization of sugars or development of carbon dioxide gases within the roast. 呈青色，草味和干草状味道。Green, grassy and hay-like taste. 以上所有 All of above 问题编码 Question ID 100009198286 以下陈述是对还是错？识别烘焙咖啡的最佳方法是从视觉上而不是通过杯测来品尝。 Is the following statement true or false? The best way to identify baked coffees is visually rather than by cupping for taste. 对 True 错 False	Alstrup, J., SCA News, 25 Magazine, <i>Modulating Perception</i>

3.05.02 质量控制方法 论 Quality Control Methodology	1. 识别生产方案中的偏差 Identify deviations in production protocol	内外杯测和三角杯测是识别偏差的有用方法。 In & out cupping and triangulation are useful methodologies to identify deviations. 在做出与新的烘焙特性和新产品发展有关的决策时，识别偏差很有帮助。 Identifying deviations is helpful when making decisions related to new roast profile and new product development.	在实操考试中考核 Tested on Practical Exam.	SCA, <i>Flavor Wheel</i>, [English], PDF SCA, CSP Sensory Intermediate Lingle, Ted. <i>The Basics of Cupping Coffee</i>, SCA
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3.06 | 栏目|批发和零售业务| Section | WHOLESALE AND RETAIL BUSINESS

2 主题

主题 Topic	子主题 Sub-topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.06.01 客户偏好 Customer Preference		1. 了解如何评估消费者的偏好 Understand how to evaluate consumer preferences	特别提到 KANO 模型分析（客户之声方法论）。 Specifically mention KANO model analysis (voice of the customer methodology).	授课，不考。 Taught, not tested.	https://www.productplan.com/glossary/kano-model/
3.06.02 价格计算 Price Calculations	3.06.02.01 烘焙咖啡的成本 Costs of Roasted Coffee	1. 了解如何计算一公斤烘焙/包装咖啡的成本 Understand how to calculate the cost of a kilogram of roasted /packaged coffee	AST 应该确保涵盖以下实例： The AST should make sure to cover instances of: - 计算生产一单位咖啡的成本 Calculating the cost of producing a unit of coffee - 在价格计算中管理烘焙损失，并Manage roast loss in price calculation 18% - 拼配策略以用于调整单位咖啡的生产成本 Blending strategies to adjust the cost of producing a unit of coffee	问题编码 Question ID 100009198287 假设以下单个项目的价格 Assume the following prices on individual items: -咖啡生豆的成本：每公斤 7 块 -带阀食品级咖啡袋：每个 0.4 块 -2 个标签（袋子两边各贴一个标签）：每个 0.01 -耗油量/千克咖啡：每公斤 0.01 -其他资金消耗/每公斤咖啡：每公斤 0.01 - Cost of green coffee: 7 per kg - Valve food grade coffee bag: 0,4 per unit - 2 labels (one label on each side of the bag): 0,01 per unit - Gas consumption/kg of coffee: 0,01 per kg - Financing/kg of coffee: 0,01per kg 卖出 1 公斤烘焙咖啡的成本是多少？ What would the cost be to sell 1 kg of roasted coffee? 8,70 /kg 9,50 /kg 8,00 /kg 10,15 /kg	

			问题编码 Question ID 100009198288 咖啡生产的哪个方面通常不算在生产一袋咖啡的成本中？ What aspect of coffee production is NOT <u>commonly</u> figured into the cost of producing a bag of coffee? 人工 Labor 燃料 Fuel 咖啡生豆成本 Green Coffee Cost <u>网站维护 Website Maintenance</u>	
	3.06.02.02 利润率 Profit Margins	1. 了解利润率计算中通常涉及的参数 Understand parameters that generally are involved in profit margin calculations	问题编码 Question ID 100009414177 在某些国家/地区，咖啡批发价预计会包含哪些额外费用或支出？选择两 (2) 个正确答案。 In some countries, what additional costs or expenses are expected to be included in the wholesale coffee price? Select the two (2) correct answers. 咖啡机 Espresso machine <u>咖啡研磨机</u> <u>Coffee grinders</u> <u>咖啡师培训</u> <u>Barista training</u> 营销支持 Marketing support 辅助的 Ancillary	

				问题编码 Question ID 100009414178 批发咖啡的利润率计算通常不属于以下哪一项 Which of the following is NOT commonly part of profit margin calculations for wholesale coffee? 出租 Rent 效用 Utility 保险 Insurance 设备折旧 Equipment depreciations 工资单 Payroll 品牌商品 Branded merchandise	
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	3.06.02.03 与年度营业额计 算相关的批次大 小要求 Batch Size Requirements Related to Yearly Turnover Calculations	1. 计算特定业务场 景所需的烘焙机大 小 Calculate the size of the roaster needed for a specific business scenario	问题 9198289 的计算: Calculation for Question 9198289: • 每天总计 6.5 小时的烘焙 时间 6.5 hours total roasting time per day • 每小时 3 批 3 batches per hour • 每天 19.5 批 19.5 batches per day • 每周 97.5 批次 97.5 batches per week • 每年 5070 批 5070 batches per year • $100000/5070 = \text{每批}$ 19.72 千克 $100000/5070 = 19.72 \text{ kg per}$ batch • $19.72/0.8 = 24.65$ 25 公斤烘焙机 a 25 kg roaster	问题编码 Question ID 100009198289 烘焙商希望他们的公司每年烘焙 100,000 公斤咖啡生 豆, 该公司只有一名员工作为他们唯一的烘焙师。 A roaster desires their company to roast 100,000 kg of green coffee per year. The company has one employee as their only roaster. <i>假设 Assume:</i> - 烘焙机只能烘焙标记容量的 80 %。 <i>The roaster can only roast 80% of marked capacity.</i> - 1 小时的机器预热和 0.5 小时的冷却时间被视为烘焙 机的工作时间。 <i>1-hour machine warmup and 0.5-hour cool-down are</i> <i>counted as work time for the roaster.</i> - 平均每小时烘焙 3 批。 <i>3 batches roasted per hour is average.</i> - 每天工作 8 小时, 每周工作 5 天。 <i>The roaster works 8 hours per day, 5 days per week.</i> - 假设增长率为 0 %。 <i>Assume 0% growth.</i> 满足所需烘焙量所需的最小咖啡烘焙机是几公斤? What is the minimum size coffee roaster that is needed to meet the desired roasting capacity? 10 公斤烘焙机 10 kilo roaster 15 公斤烘焙机 15 kilo roaster 25 公斤烘焙机 25 kilo roaster 35 公斤烘焙机 35 kilo roaster	
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		问题 100009198290 的计算： Calculations for Question 100009198290: • 每天总计 6.5 小时的烘焙时间 6.5 hours total roasting time per day • 每小时 3 批 3 batches per hour • 每天 19.5 批 19.5 batches per day • 每周 58.5 批 58.5 batches per week • 每年 3042 批 3042 batches per year • 75000/3042 =每批次 24.6 千克 75000/3042= 24.6 kg per batch • 24.6 / 0.8 =至少 30.8 千克烘焙机 24.6/0.8= 30.8 kg roaster minimum	问题编码 Question ID 100009198290 烘焙商希望他们的公司每年烘焙 100,000 公斤咖啡生豆，该公司只有一名员工作为他们唯一的烘焙师。 A roaster desires their company to roast 100,000 kg of green coffee per year. The company has one employee as their only roaster. The company has 1 employee as their only roaster. 假设 Assume: - 烘焙机只能烘焙标记容量的 80 %。 The roaster can only roast 80% of marked capacity. - 1 小时的机器预热和 0.5 小时的冷却时间被视为烘焙机的工作时间。 - 1-hour machine warmup and 0.5-hour cool-down are counted as work time for the roaster. - 平均每小时烘焙 3 批。 - 3 batches roasted per hour is average. - 每天工作 8 个小时，每周工作 3 天。 - The roaster works 8 hours per day, 3 days per week. - 假设增长率为 0 %。 - Assume 0% growth. 满足所需烘焙量所需的最小咖啡烘焙机是几公斤？ What is the minimum size coffee roaster that is needed to meet the desired roasting capacity? 50 公斤烘焙机 50 kilo roaster 15 公斤烘焙机 15 kilo roaster 25 公斤烘焙机 25 kilo roaster 35 公斤烘焙机 35 kilo roaster	
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3.07|栏目|烘焙厂管理 | Section | ROASTERY MANAGEMENT

5 主题

主题 Topic	子主题 Sub-topic	目标 Objectives	AST 注意事项 AST Notes	在线笔试题 Online Written Exam Questions	参考资料 References
3.07.01 烘焙生产设计与优化 Roastery Production Design and Optimization		1. 生产和库存控制技术的基础知识 Basic knowledge of production and inventory control techniques	AST应该讨论如何设计和操作针对最大运行时间和最满意的客户（瑕疵最少）的客户喜好和需求的产品。 AST should discuss how to design and operate a production that is aimed toward the preferences and demands of customers with maximum uptime and most satisfied customers (fewest defects). AST 确保涵盖：精益管理和PEMME，以及其他生产和库存控制技术。AST to ensure to cover: LEAN management and PEMME alongside other production and inventory control techniques.	问题编码 Question ID 100009198291 缩写 PEMME 在 LEAN 术语中代表什么？ What does the abbreviation PEMME stand for in LEAN terminology? 包装，设备，方法，制造，环境 Packing, Equipment, Method, Manufacture, Environment 过程，设备，测量，材料，环境 Process, Equipment, Measurement, Material, Environment 人，设备，方法，材料，环境 People, Equipment, Method, Material, Environment	Morgan, J. & Brenig-Jones, M., <i>Lean Six Sigma for Dummies</i> , 23-41
				问题编码 Question ID 100009198292 选择所有 PEMME 分析示例： Select all examples of PEMME analysis: 接收生豆 Green bean reception 包装区域 Packing area 烘焙过程 Roasting process 实验室测量 Laboratory measurements 以上所有 All of the above	
3.07.02 供应链管理 Supply Chain Management		1. 了解在客户需求时始终有可用的产品的需求 Understand need to always have product available when demanded by customer	购买计划要求 Purchase planning requires: -了解咖啡供应的季节性 Knowledge of seasonality of coffee availability	授课，不考。 Taught, not tested.	

			-能够预先计划以确保从收获到收获所需的咖啡生豆量 Ability to plan ahead to secure amounts of green coffee needed from harvest to harvest -能够计划购买其他生产物品，例如袋子，杯子，糖浆，标签等。Ability to plan purchase of other production items like bags, to go cups, syrups, labels, etc.		
3.07.03 监控烘焙过程 Monitoring Roasting Process	3.07.03.01 颜色测量的变化 Variations of Color Measurements	1. 展示测量不同色值变化并计算结果的能力 Demonstrate ability to measure different roasted color variations and calculate results		在实操考试中考核 Tested on Practical Exam.	www.coffee-mind.com/variation-analysis/
	3.07.03.02 烘焙记录软件 Roast Log Software	1. 展示使用烘焙日志软件作为质量控制检查的一部分的能力 Demonstrate ability to use roast log software as part of quality control checks		在实操考试中考核 Tested on Practical Exam.	

3.07.04 拼配 Blending	3.07.04.01 拼配成分数 Number of Blend Components	1. 了解太多或太少的拼配成分的利弊 Understand the benefits and drawbacks of too many or too few blend components	较少的成分拼配 Blends with fewer components: - 通常, 生产更简单 Generally, simpler to produce. -由于季节性或取得性, 可能会发生更明显的变化。May suffer more noticeable changes due to seasonality or availability. 许多成分拼配 Blends with many components: -通常, 受季节性因素或一种或多种成分的取得性的影响较小 Generally, less affected by seasonality or availability of one or more components -拼配的生产管理可能变得太困难 Production management of the blend can become too difficult -难以保持一致的结果 Consistent results are difficult to maintain 当拼配的百分比太小时, 在每个冲煮把手或冲煮滤杯获得每种成分一致性正确百分比的可能性就很小。When the percentage of a blend becomes too small, the likeliness of getting the correct percentage of each component in every portafilter or brew-basket is small. 可以重新制作或融合回去作为调整选项。Re-work or blend back as an option to adjust.	问题编码 Question ID 100009198293 创建浓缩咖啡拼配时, 避免使用太多拼配成分的一个很好的理由是什么? What could be a good reason to avoid using too many blend components when creating an espresso blend? 烘焙咖啡的发展不足可能是一个问题 Underdevelopment of the roasted coffee can be a problem. 该拼配的生产可能很昂贵 The blend can be expensive to produce. <u>拼配的生产管理可能太困难了 Production management of the blend can be too difficult.</u> 季节性和切换成分可能会过度影响风味 Seasonality and switching components can over influence the flavor. 问题编码 Question ID 100009198294 在饮料组合中简单拼配 (由 2 种不同成分组成) 有什么缺点? What is a disadvantage of having a simple blend (made up of 2 different components) in the drink portfolio? <u>拼配可能会受季节性或一或两种成分的取得性而影响 The blend may be affected by seasonality or availability of one or both components.</u> 拼配生产可能很复杂 The blend can be complex to produce. 拼配生产太昂贵 The blend can be too expensive to produce. 简单拼配料没有缺点 There is no disadvantage to simple blends.	Illy & Viani, <i>Espresso Coffee</i>, 141-142 Roast Magazine, <i>The Book of Roast</i>, 357-358 Folmer, Britta, and Imre Blank. <i>The Craft and Science of Coffee</i>, 268
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				问题编码 Question ID 100009198295 当其中一种成分的比例不足总比例的 5% 时，会对咖啡拼配产生什么影响？ What could be the impact on a coffee blend when one of the components represents less than 5% of the overall blend? 拼配的生产将更加复杂 The blend will be more complex to produce. 萃取物的成分可能不一致 The extraction will likely not be consistent in make-up. 萃取物可能在成品中非常一致 The extraction will likely be very consistent in make-up. 拼配的口味会更加复杂 The blend will be more complex in taste.	
	3.07.04.02 生拼/熟拼 Pre/post Blending	1. 了解生拼和熟拼的优缺点 Understand the benefits and drawbacks of pre/post blending	后掺混提供了最佳烘焙各组分的机会，但具有额外的生产步骤的成本。 Post blending provides the opportunity to roast each component optimally but has the cost of extra production steps.	问题编码 Question ID 100009198296 熟拼的优势是什么？ What is an advantage of post-blending coffee? 最不困难的生产方式 Least difficult method for the production team. 将烘焙次数降至最低 Minimizes the number of production roasts 最大化拼配产品 Maximizes the blend offerings 避免过度生产 Avoids overproduction	

			请记住，烘焙前拼配和烘焙后拼配均可产生优质的饮料。 Remember that pre-roast blending and post- roast blending both produce quality drinks.	问题编码 Question ID 100009198297 生拼的优势是什么？ What is an advantage of blending before roasting? 最终产品在浓缩咖啡方面将更加一致 The final product will be more consistent in terms of espresso extraction. <u>生产团队可以更快更省力地工作 It is faster and less effort on the part of the production team.</u> 无论咖啡的来源或处理法为何，最终的烘焙产品将具有一致的颜色。 Independently of coffee origin and or process, the final roasted product will have a consistent color. 每个咖啡都以最佳烘焙曲线焙製 Each coffee will develop its optimum profile. 问题编码 Question ID 100009198298 假设一个烘焙师在咖啡烘焙厂中工作，而他们没有单独的混豆机。烘焙机每天稼动 12 小时，轮班制操作。哪种拼配法是最经济的解决方案？ Assume a roaster is working in a coffee roastery where they do not have access to a separate blending machine. The roaster is roasting for 12 hours per day with split shifts on the machine. Which is the most economical solution for blending? <u>预烘焙拼配（生拼）</u> <u>Pre-roast blending</u> 烘焙后拼配（熟拼） Post-roast blending	
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3.07.05 烘焙度对清洁和维护的影响 Roast Degree's Effect on Cleaning and Maintenance		1. 了解烘焙程度如何影响清洁和烘焙维护的频率 Understand how roast degree affects the frequency of cleaning and roaster maintenance	在深烘焙时，清洁变得更加频繁（烘焙机，管道，桶，漏斗等）。 When roasting darkly, cleaning becomes more frequent (roaster, pipes, buckets, hoppers, etc.).	问题编码 Question ID 100009198299 如果一家公司主要烘焙深烘焙咖啡（即二爆及以后），并且频繁烘焙（每天 20 批以上），那么烘焙机的清洁时间表是多久一次？ If a company is roasting primarily dark roasted coffees (i.e. to second crack and beyond) and is roasting frequently (20+ batches per day), what is the frequency of the roaster's cleaning schedule? <u>每月一次</u> <u>Monthly</u> 每季一次 Quarterly 一年两次 Twice a year 每年一次 Yearly 问题编码 Question ID 100009198300 主要采取浅烘焙，与深烘焙相比，维护保养的频率应更高或更少？ When primarily light roasting, is the maintenance expected to be more or less frequent compared with darker roasting? 更频繁 More frequent <u>较不频繁</u> <u>Less frequent</u> 烘焙度不影响清洁时间表 Roast degree does not affect the cleaning schedule.	
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5. 必备的 SCA 培训文档 Essential SCA Training Documents

- Certification System Materials
- In and Out Base sheets
- Practical Exam Base Sheets
- Profile Log Professional (C and F)
- Roast Color Correlation
- Simplified Roast Evaluation
- SCA Cupping Form
- SCA Coffee Taster's Flavor Wheel (English)

所有文档可以在官网 AST 门户中的烘焙课程与考试资料文件夹中找到 All documents are available at the AST Portal under Curriculum and Exams/Roasting.

6. 所需设备和用品清单 Required Equipment and Supplies List

可在 AST Portal 的“资源/场地要求”下找到。SCA 美国或英国商店中可以购买的物品均会提供链接。

Available at the AST Portal under Resources/Venue Requirements. Any items available in the SCA US or UK store are noted and a link directly to the store is provided.

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8. 推荐阅读 Recommended Reading

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